

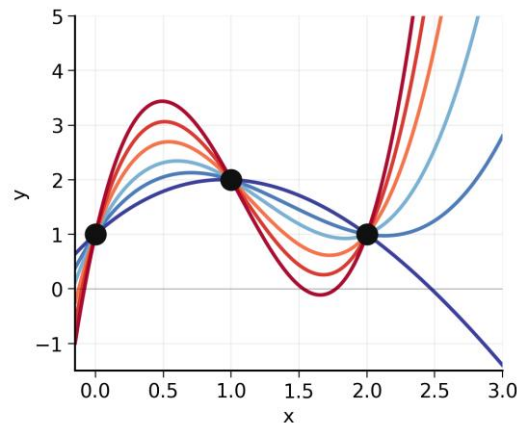
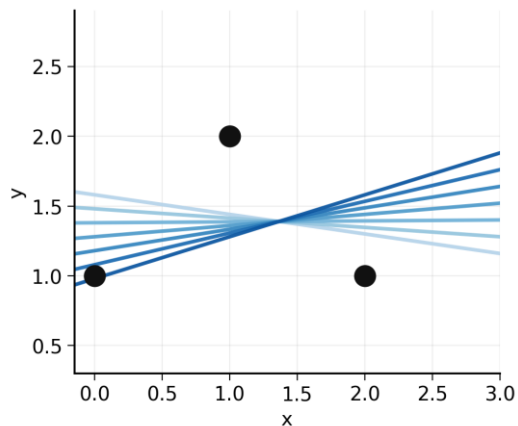
Loss Landscape Analysis

Understanding underlying NN behavior

Bayesian Neural Network

- **What is Bayesian Neural Network?**

- Make reliable predictions based on uncertainty!
- Data uncertainty => irreducible
- Model uncertainty => reducible

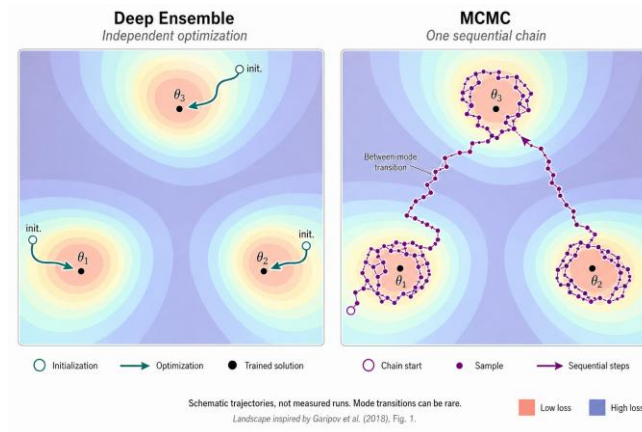


How BNN works?

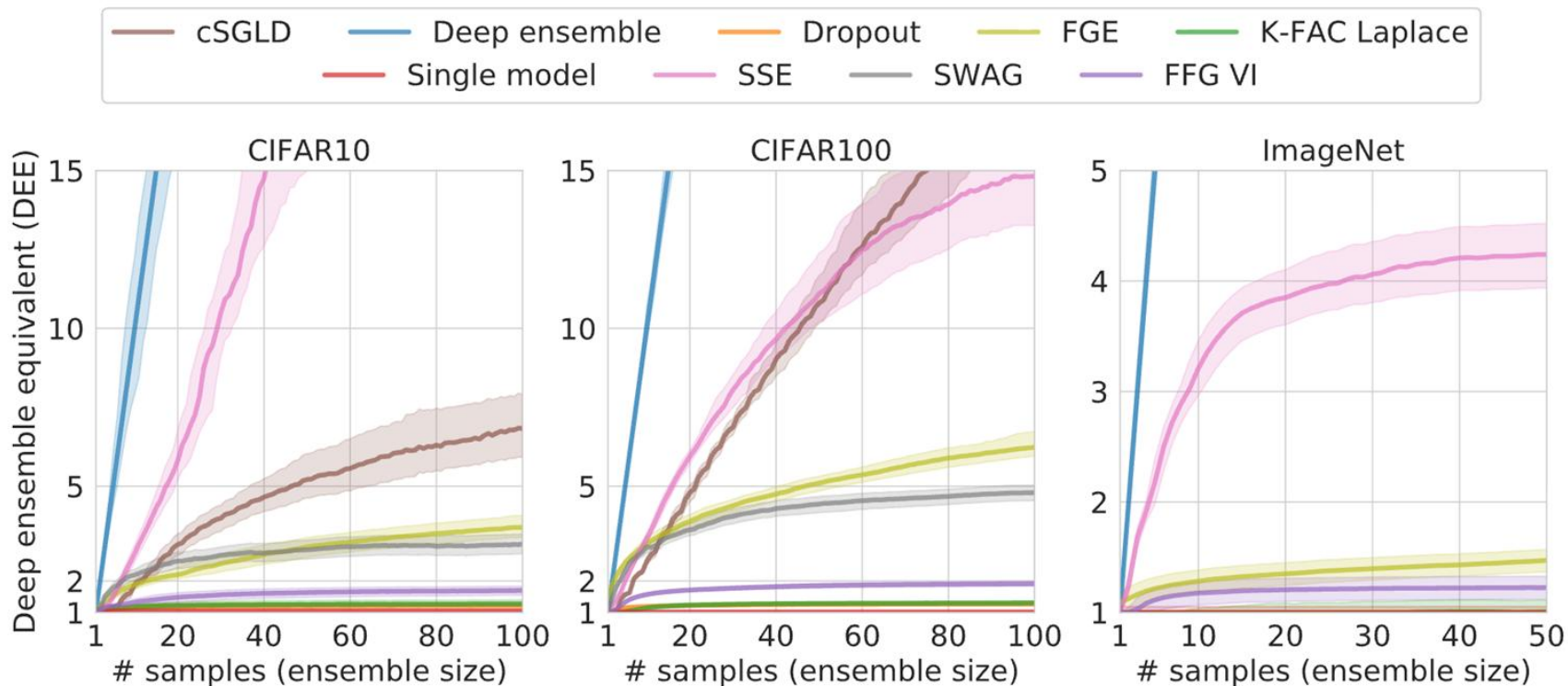
- Bayesian Model Averaging
 - $p(y|x, D) = \int p(y|x, \theta)p(\theta|D)d\theta$
 - Here, θ is the NN parameter and $p(y|x, \theta)$ is the model prediction.
- Almost impossible to compute in closed form... approximation?
$$p(y|x, D) \approx \frac{1}{N} \sum_{i=1}^N p(y|x, \theta_i)$$
- Theoretically grounded =>MCMC
- Crude approximation => Deep Ensemble

MCMC vs Deep Ensemble

- **MCMC** => Sequentially moving around the loss surface and collect the parameters
- **Deep Ensemble** => Using random initialization and find the mode near initialization
- Problem?
 - Usually 4~8 particles
 - Deep Ensemble > MCMC!!

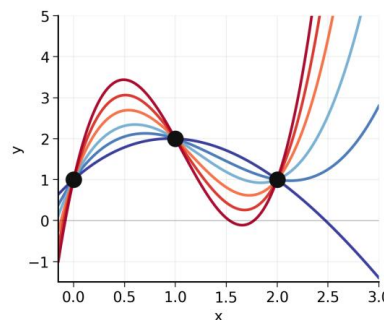
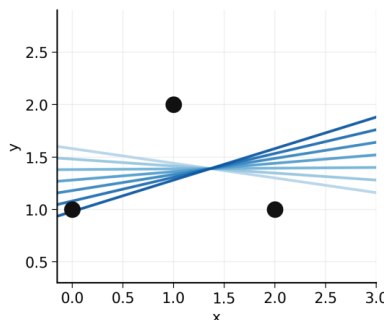


MCMC vs Deep Ensemble



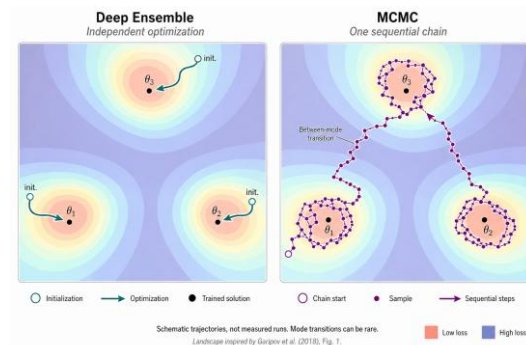
MCMC vs Deep Ensemble

- **Why?** Sequential nature of MCMC and the lack of particles...



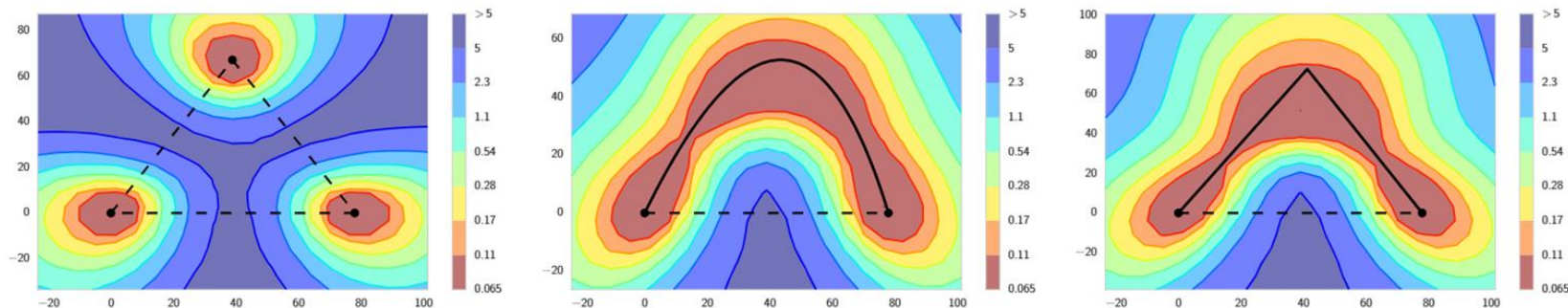
- Main issue?
 - Goal: Make MCMC that can efficiently moving around the mode

How? **Finding the low loss path between modes**



Mode Connectivity (Garipov et al., 2018, Draxler et al., 2018)

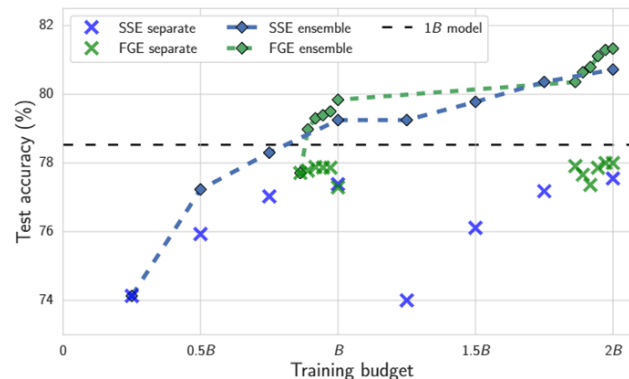
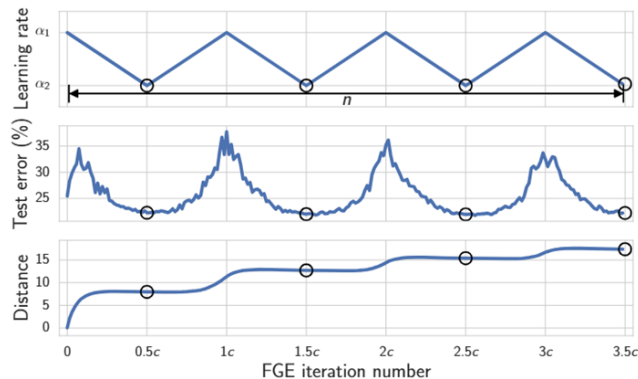
- **Fail to find the linear path... However?**



- We can find the path even with simple bezier curve!!
- Often $\theta_{0.5}$ is better than θ_0 or θ_1 !
- NICE!!!!

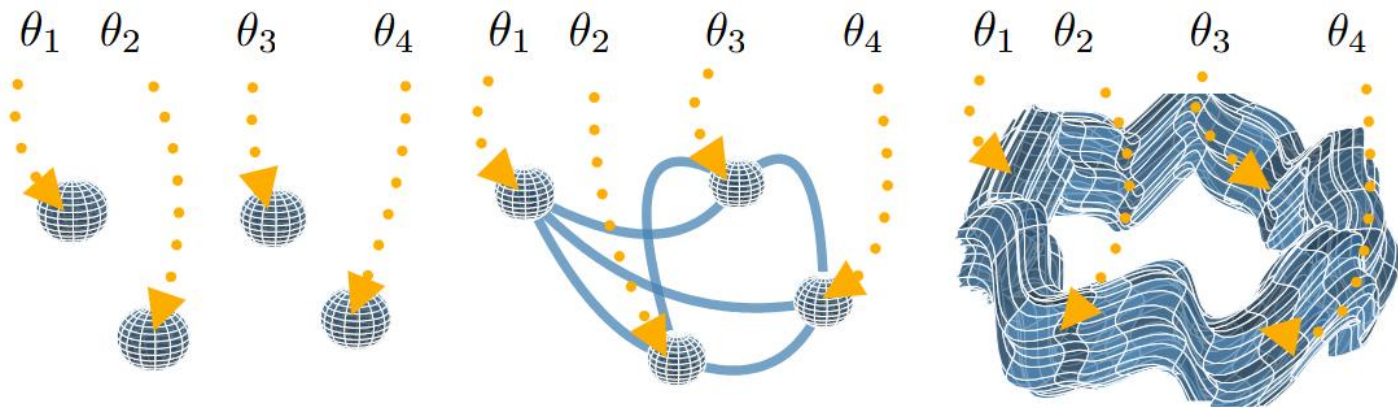
Fast Geometric Ensembling (Garipov et al., 2018)

- Let's move toward low loss curve!!



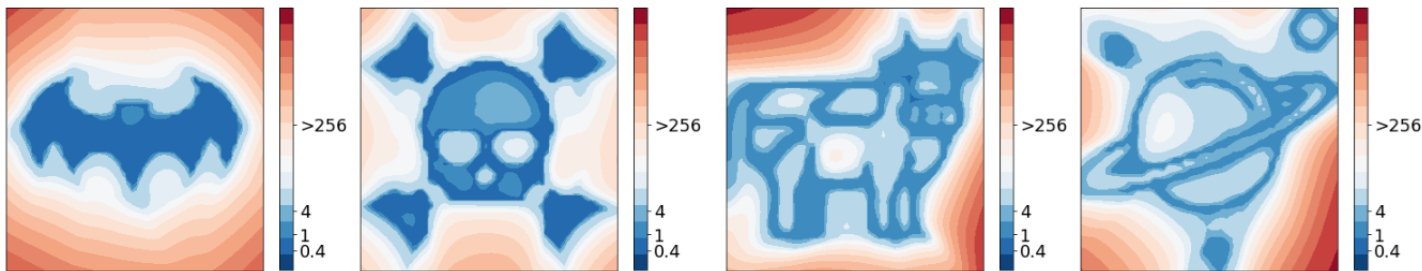
Only line? (Benton et al., 2021)

- Move towards the curve exactly is quite hard...
- Can we find some space with some amount of volume?



Even more??? (Ivan et al, 2019)

- **Bring any pattern then we can always find that pattern in the loss landscape!**



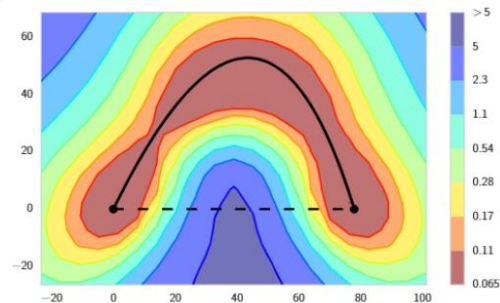
(a) Loss surface on FashionMNIST dataset

(b) Loss surface on CIFAR10 dataset

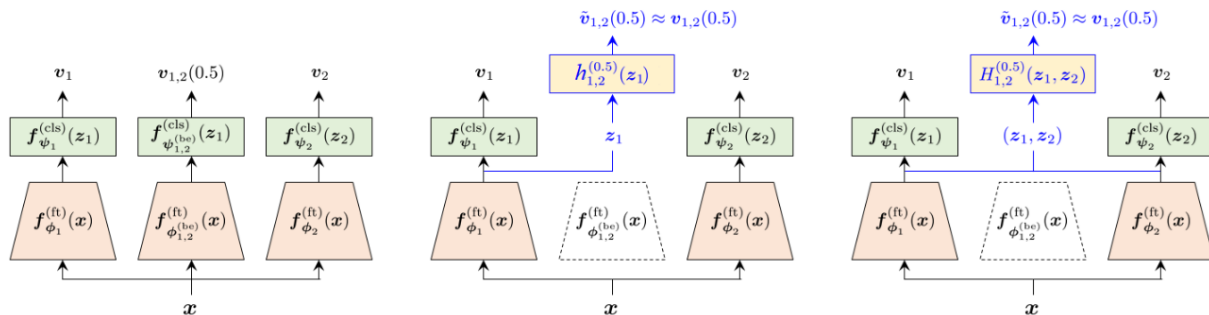
- **Central message: parameter space is so huge, so that we can easily find some patterns of loss landscape**

My own trial (ICML 2023)

- Smooth path? Smooth function evolving?

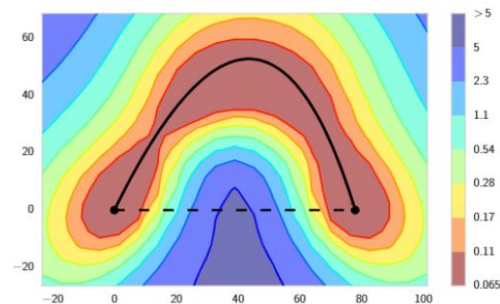


- Let's efficiently imitate the output of $\theta_{0.5}$ or θ_1



My own trial but failed...

- Can we generate θ_1 using θ_0 with Diffusion process or any generation models?
- Not easy...

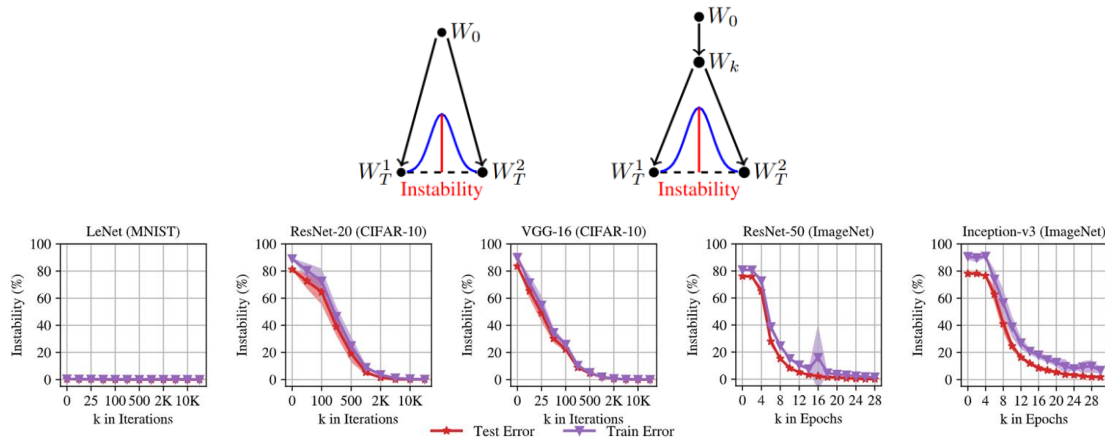


Really really no linear connectivity?

- **Nonlinear pathway still makes lot of problems (Even with space)**
- It is hard to find the pathway itself
- It is hard to follow the pathway without learning the shape or predefine the shape.
- People tried to analyze about linear connectivity:
 - Is it really impossible to find linear connectivity?
 - When does linear connectivity appears?
 - What cause the nonlinear pathway?

When does linear connectivity success? (Frankle et al., 2020)

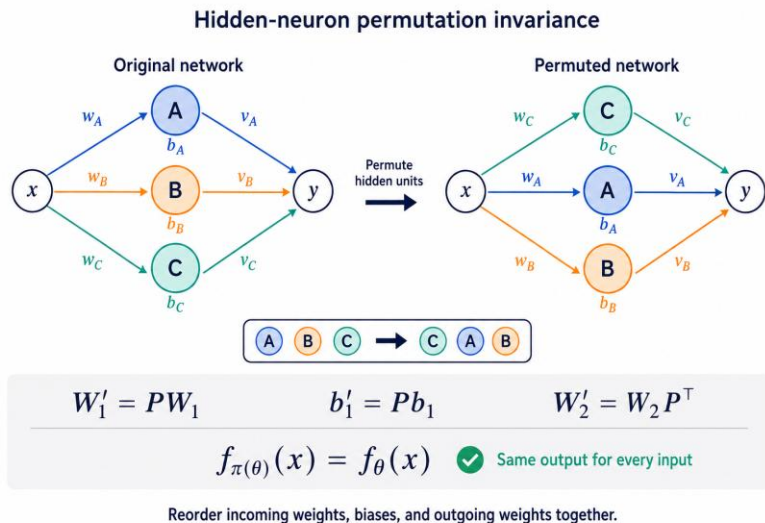
- Deep Ensemble => random initialization
- If our particles share some learning trajectories?



- Shared something or at least some aligned structure

Alignment? (Entezari et al., 2022)

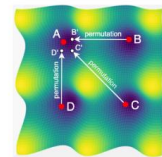
- If alignment is important, can we align our particles using permutations?
 - Permutation does not make neural network model different!



Alignment? (Entezari et al., 2022)

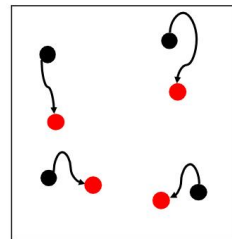
- Can we transfer one parameter from the other local minima to the same local minima with permutations?

$$\min_{\pi} \epsilon \left(\frac{\theta_0 + \pi(\theta_1)}{2} \right)$$

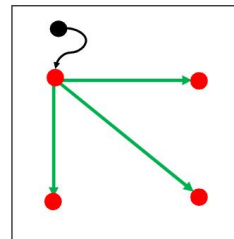


Real World

Our Model



$\approx$

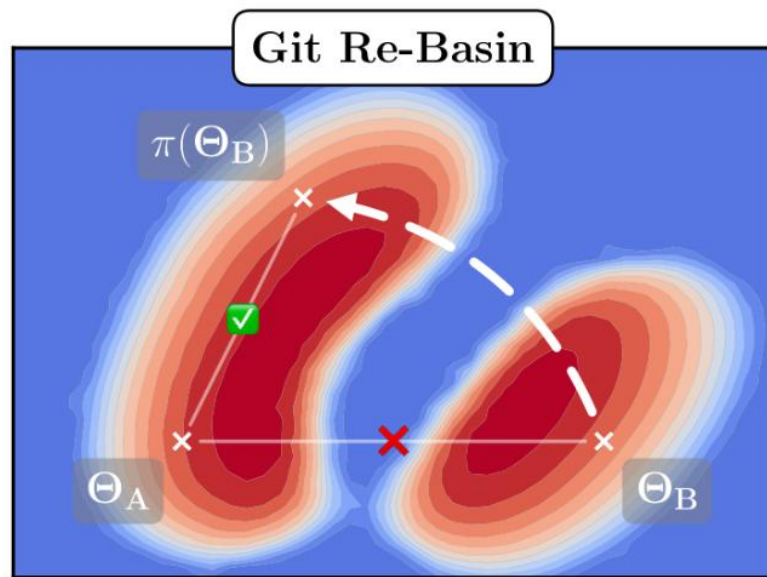


● Random Init.
● Final parameters

↪ SGD Path
→ Random Perm.

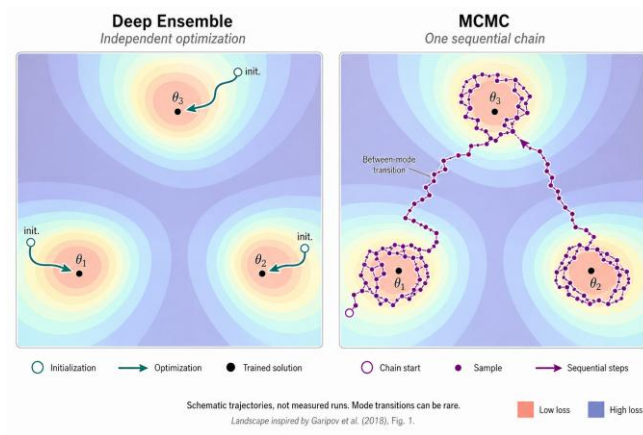
Fancier way? (Ainsworth et al., 2023)

- Find this permutation using activation matching and weight space matching, etc.
- More easy to find good matching
- !! Shows some counterexamples...



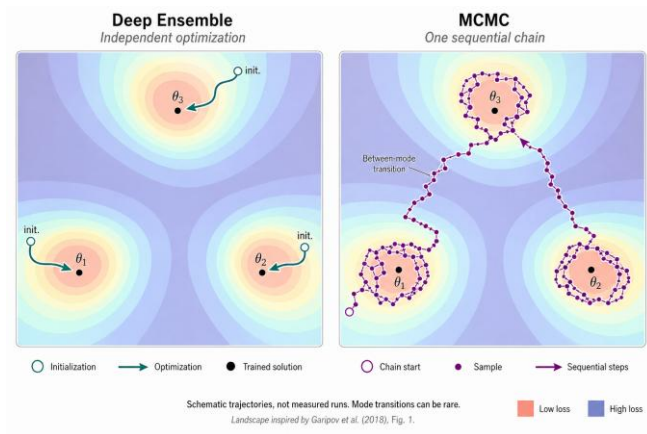
Hope... Expectation...

- If we can gather all the solutions or some portion of the solutions in convex hull, then Bayesian Model Averaging can be done more accurately only using a good local approximation!
- We already have lots of Variational Inference methods@@@@



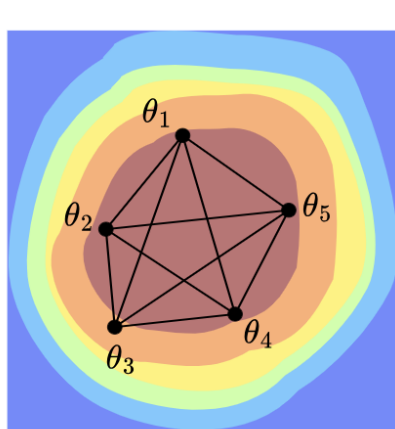
Hope... Expectation...

- Also, we can find lots of good other solutions by seeking near the solution.

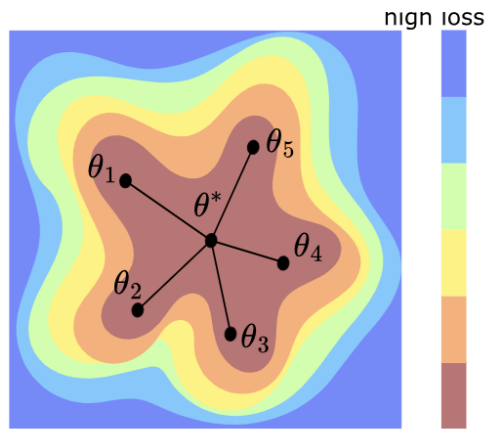


No free lunch (Sonthalia et al., 2024)

- Sadly one paper empirically shows that it is quite hard (maybe impossible) to gather particles into the some convex hull.
- One recent paper theoretically shows this phenomenon as well.



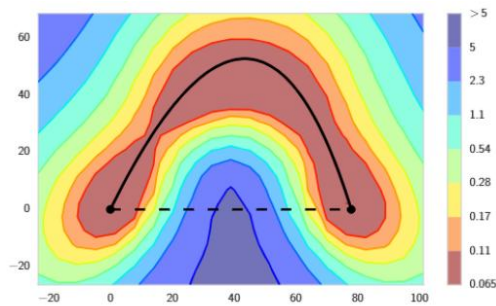
Convexity conjecture (Entezari et al.)
holds only for wide networks.



Star domain conjecture (ours)
holds even for narrower networks.

However weight space averaging_(Ito et al., 2025)

- Weight averaging usually gives performance improvement.
- Even with similar weights (s.a. weights in the same learning trajectory), it gives noticeable improvement.
- What if we average totally different weights?
- !! Not sure it really helps...



Algorithm 1 STE-MM

Require: Models $\theta_1, \theta_2, \dots, \theta_n$, the number of epochs n_e

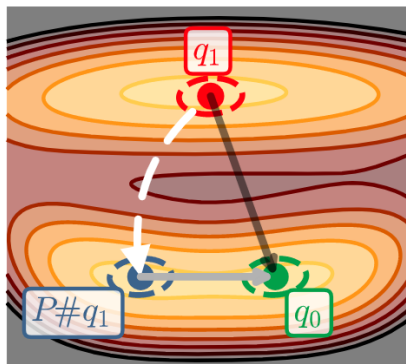
Ensure: Permutations $\pi_1, \dots, \pi_{n-1}$

```
1:  $\hat{\theta}_i \leftarrow \theta_n$  for all  $i \in [n-1]$   $\triangleright$  Initialize dummy parameters  $\hat{\theta}_i$ .
2: for  $e = 1 \dots N_e$  do
3:    $\pi'_i \leftarrow \text{WM}(\hat{\theta}_i, \theta_i)$  for all  $i \in [n-1]$ 
4:    $(\lambda_i)_{i=1}^n \leftarrow \text{UniformDist}((0, 1))$ 
5:    $\lambda_i \leftarrow \lambda_i / \sum_j \lambda_j$  for all  $i \in [n]$ 
6:    $\text{loss} \leftarrow \mathcal{L}(\sum_{i=1}^{n-1} \lambda_i \pi'_i(\theta_i) + \lambda_n \theta_n)$ 
7:   for  $i = 1 \dots n-1$  do
8:      $\text{grad}_i \leftarrow \text{GetGradient}(\text{loss}, \pi'_i(\theta_i))$ 
9:      $\hat{\theta}_i \leftarrow \text{Update}(\hat{\theta}_i, \text{grad}_i)$ 
10:  end for
11: end for
12:  $\pi_i \leftarrow \text{WM}(\hat{\theta}_i, \theta_i)$  for all  $i \in [n-1]$ 
```

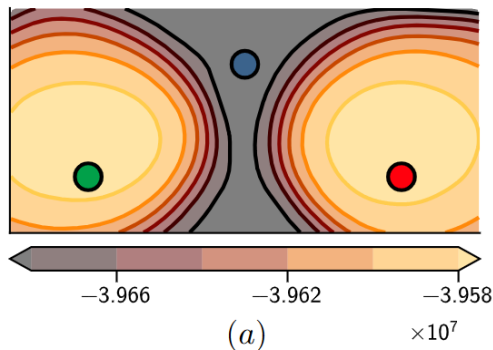
Another direction in BNN_(Rossi et al., 2023)

- Let's directly align posterior approximations!!
- Anyway get some improved BMA approximations :)

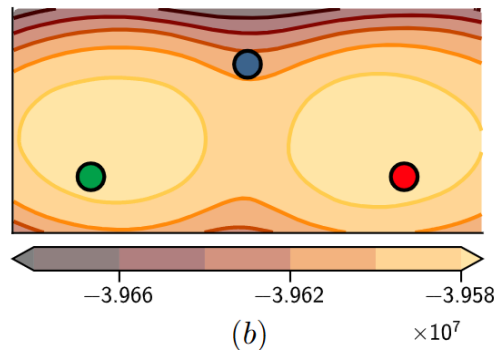
Log-posterior for CIFAR10



Posterior without alignment



Posterior with alignment



(a)	(b)
q_0	q_0
q_τ	$P\#q_\tau$
q_1	$P\#q_1$

Note: $\tau = 0.5$

There are lots of possible research directions

- **For example:**
 - **Quantization => Maximize the volume size of low loss area so that it can cover the quantized particle**
 - **Pruning => Iterative pruning and linear mode connectivity**

Other

- **Sharpness or Flatness in local minima**
 - SAM
- **Asymmetric valley**
 - SWA
- **Learning Theory**
 - Lazy learning regime (NTK)
 - Feature learning regime (μ -parametrization)

Q&A